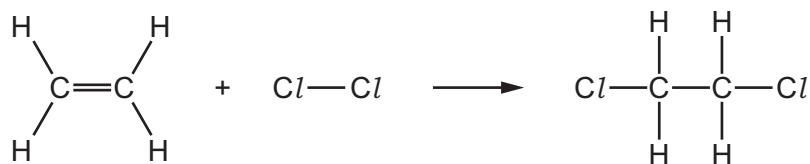


(ii) The chemical equation for this reaction can be represented as shown.



The energy change for the reaction is -180 kJ/mol .

Use the bond energies in the table to calculate the bond energy of a $\text{C}-\text{Cl}$ bond, in kJ/mol .

bond	C-H	C=C	Cl-Cl	C-C
bond energy in kJ/mol	410	610	240	350

Use the following steps.

step 1 Calculate the energy needed to break bonds.

energy needed to break bonds = kJ

step 2 Use your answer in **step 1** and the energy change for the reaction to determine the energy released when bonds are formed.

energy released when bonds form = kJ

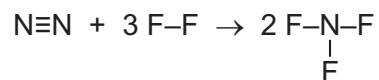
step 3 Use your answer in **step 2** and bond energy values to determine the energy of a $\text{C}-\text{Cl}$ bond.

bond energy of a $\text{C}-\text{Cl}$ bond = kJ/mol
[4]

[Total: 11]

(b) Nitrogen reacts with fluorine to form nitrogen trifluoride, NF_3 .

(i) The chemical equation can be represented as shown.



Some bond energies are shown in the table.

bond	bond energy in kJ/mol
$\text{N}\equiv\text{N}$	945
$\text{F}-\text{F}$	160
$\text{N}-\text{F}$	300

Calculate the energy change for the reaction between nitrogen and fluorine, using the following steps:

- energy taken in to break bonds

..... kJ

- energy released when bonds are formed

..... kJ

- energy change during the reaction.

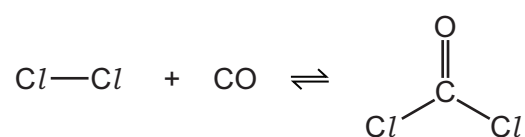
..... kJ/mol
[3]

(ii) Use your answer to (i) to deduce whether this reaction is endothermic or exothermic. Explain your answer.

.....

..... [1]

(d) The reaction between chlorine and carbon monoxide can be represented as shown.



When one mole of chlorine reacts with one mole of carbon monoxide, 230 kJ of energy is released.

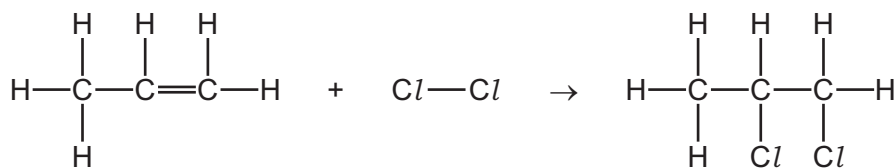
Some bond energies are shown in the table.

bond	bond energy in kJ/mol
Cl-Cl	240
C=O	745
C-Cl	400

Use the information to calculate the energy of the bond between the C and the O in carbon monoxide, CO.

bond energy in carbon monoxide, CO = kJ/mol [3]

(ii) The structures of the reactants and products of this reaction are shown.



Some bond energies are shown in the table.

bond	bond energy in kJ/mol
C–C	347
C=C	612
C–H	413
C–Cl	339
Cl–Cl	242

Calculate the energy change for the reaction between propene and chlorine using the following steps.

- Calculate the energy needed to break the bonds.

..... kJ

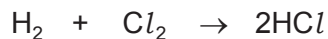
- Calculate the energy released when bonds are formed.

..... kJ

- Calculate the energy change for the reaction between propene and chlorine.

..... kJ/mol
[3]

(d) Hydrogen and chlorine react to form hydrogen chloride gas, as shown in the equation.



This equation can be represented as shown.



Some bond energies are shown in the table.

bond	bond energy in kJ/mol
H-H	436
Cl-Cl	243
H-Cl	432

Calculate the energy change for the reaction between hydrogen and chlorine, using the following steps.

- Calculate the energy needed to break the bonds.

..... kJ

- Calculate the energy released when bonds are formed.

..... kJ

- Calculate the energy change for the reaction.

..... kJ/mol
[3]

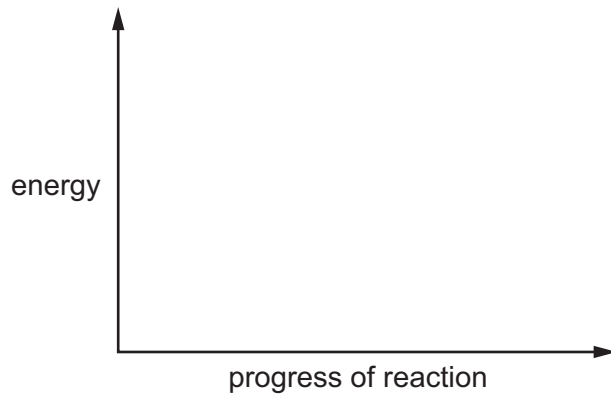
[Total: 13]

(b) The reaction between ethanoic acid and ethanol is exothermic.

Draw an energy level diagram for this reaction.

On your diagram label:

- the reactants and products
- the energy change of the reaction, ΔH .



[3]

(c) Concentrated sulfuric acid is a catalyst for this reaction.

What is meant by the term *catalyst*?

.....

..... [2]

- 2 A student investigated the temperature changes when two different metals, zinc and magnesium, reacted with aqueous copper(II) sulfate.

Three experiments were done.

Experiment 1

- A measuring cylinder was used to pour 25 cm^3 aqueous copper(II) sulfate into a polystyrene cup.
- The initial temperature of the solution was measured and the timer was started.
- The temperature of the solution was measured at 30 seconds and at 60 seconds.
- At 60 seconds, 5g of zinc powder was added to the aqueous copper(II) sulfate. The mixture was stirred with a thermometer.
- The temperature of the mixture was measured every 30 seconds for 210 seconds. The mixture was stirred continuously.

(a) Use the thermometer diagrams to record the temperatures in the table.

time / s	0	30	60	90	120	150	180	210
thermometer diagram								
temperature of mixture / °C								

[2]

Experiment 2

- Experiment 1 was repeated using 5g of magnesium powder instead of zinc powder.

(b) Use the thermometer diagrams to record the temperatures in the table.

time / s	0	30	60	90	120	150	180	210
thermometer diagram								
temperature of mixture / °C								

[1]

Experiment 3

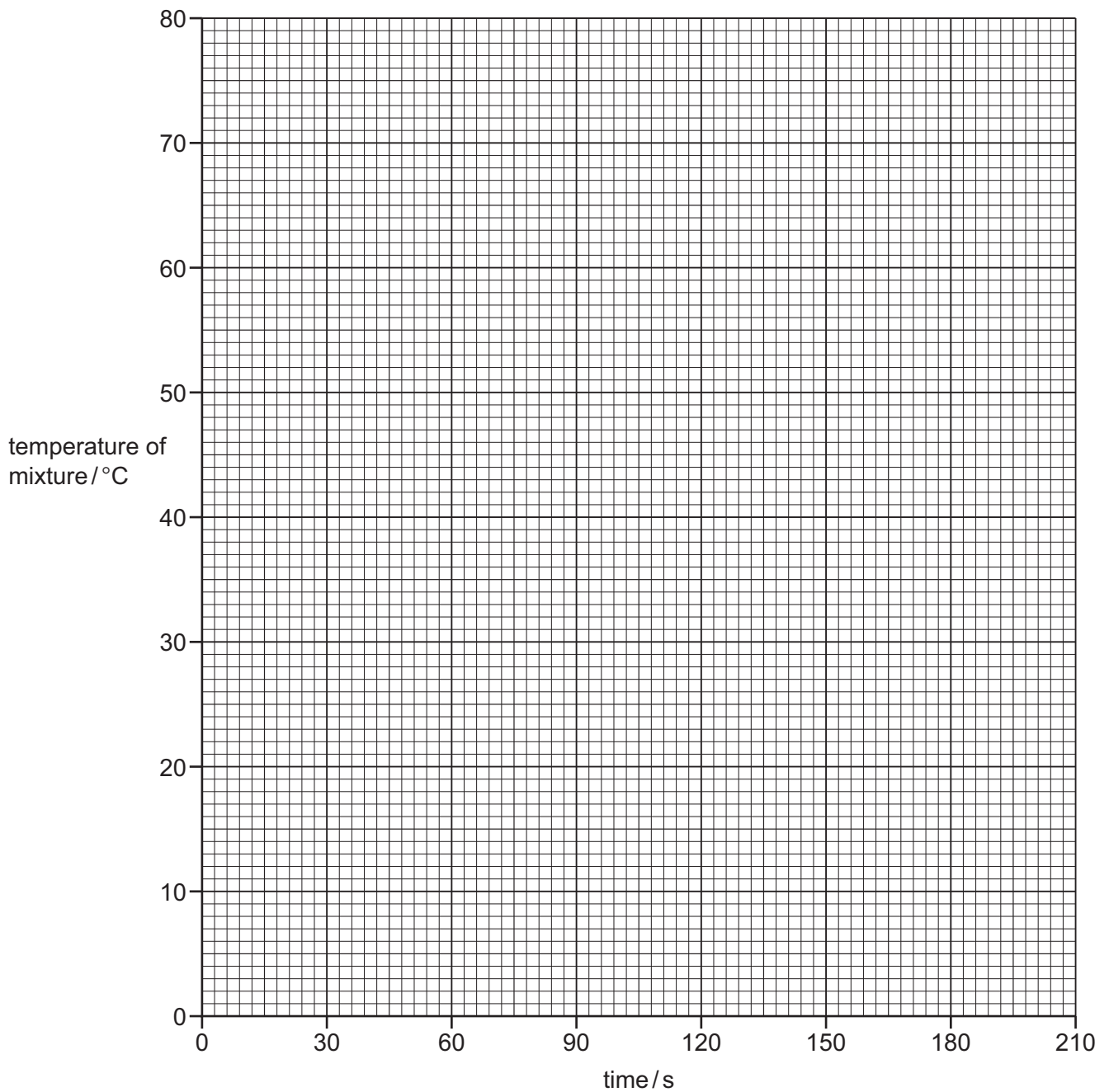
- Experiment 1 was repeated using 5g of zinc granules instead of zinc powder.

(c) Use the thermometer diagrams to record the temperatures in the table.

time / s	0	30	60	90	120	150	180	210
thermometer diagram								
temperature of mixture / °C								

[1]

- (d) Plot the results for Experiments 1–3 on the grid and draw **three** smooth line graphs. Clearly label your lines.



[3]

- (e) From your graph, deduce the temperature of the mixture in Experiment 2 after 75 seconds.

Show clearly **on the grid** how you worked out your answer.

..... °C [2]

- (f) (i) From the results, which Experiment was the most exothermic? Explain your answer.

.....
..... [2]

- (ii) Compare the rates of reaction in Experiments 1 and 3. Explain why the rates of reaction are different.

.....
.....
..... [2]

- (g) Predict the temperature of the mixture in Experiment 2 after 2 hours. Explain your answer.

.....
..... [2]



- (h) When doing the experiments, what would be the advantage of taking the temperature readings every 15 seconds?

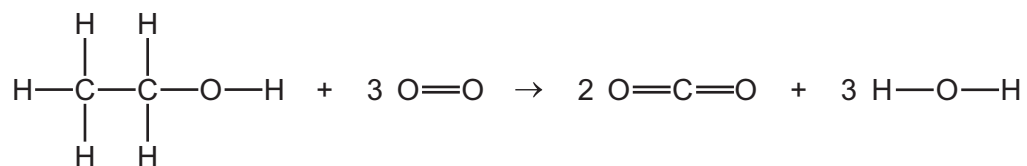
.....
..... [2]

- (i) Explain why a copper can should **not** be used in place of the polystyrene cup in these experiments.

.....
.....
..... [2]

[Total: 19]

(b) The equation for the complete combustion of ethanol is shown.



Use the bond energies in the table to calculate the energy change, in kJ/mol, for the complete combustion of ethanol.

bond	bond energy in kJ/mol
C–C	347
C–H	413
C–O	358
C=O	805
O–H	464
O=O	498

- Energy needed to break bonds.

..... kJ

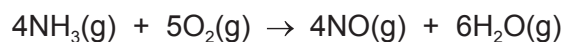
- Energy released when bonds are formed.

..... kJ

- Energy change for the complete combustion of ethanol.

energy change = kJ/mol
[3]

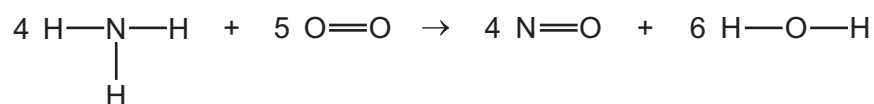
(d) Ammonia reacts with oxygen as shown.



- (i) Calculate the volume of oxygen at room temperature and pressure, in dm^3 , that reacts with 4.80 dm^3 of ammonia.

volume = dm^3 [3]

- (ii) The chemical equation for the reaction can be represented as shown.



Use the bond energies in the table to calculate the energy change, in kJ/mol , which occurs when **one** mole of NH_3 reacts.

bond	N–H	O=O	N=O	O–H
bond energy in kJ/mol	391	498	587	464

- Energy needed to break bonds.

..... kJ

- Energy released when bonds are formed.

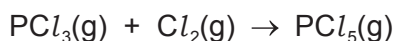
..... kJ

- Energy change when **one** mole of NH_3 reacts.

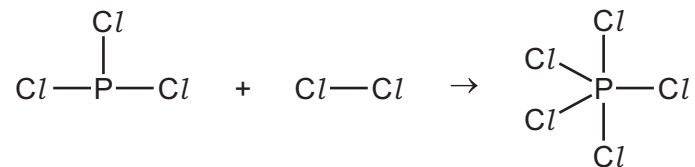
energy change = kJ/mol
[4]

[Total: 22]

- (c) Gaseous phosphorus(III) chloride, PCl_3 , reacts with gaseous chlorine to form gaseous phosphorus(V) chloride, PCl_5 .



The chemical equation for this reaction can be represented as shown.



- (i) Use the bond energies in the table to calculate the energy change, in kJ/mol, of the reaction.

bond	bond energy in kJ/mol
P-Cl	326
Cl-Cl	243

- Energy needed to break bonds.

..... kJ

- Energy released when bonds are formed.

..... kJ

- Energy change of reaction.

energy change = kJ/mol
[3]

- (ii) Deduce whether the energy change for this reaction is exothermic or endothermic. Explain your answer.

.....
..... [1]

- 2 A student investigated the temperature changes when two different solids, **N** and **O**, dissolve in water.
Two experiments were done.

Experiment 1

- Using a measuring cylinder, 30 cm³ of distilled water was poured into a polystyrene cup.
- The initial temperature of the distilled water was measured.
- Solid **N** was added to the distilled water, a timer started and the mixture was stirred with a stirring thermometer.
- The temperature of the mixture was measured every 30 seconds for three minutes (180 seconds).

(a) Use the thermometer diagrams to record the temperatures in the table.

time / s	0	30	60	90	120	150	180
thermometer diagram							
temperature of mixture / °C							

[2]

Experiment 2

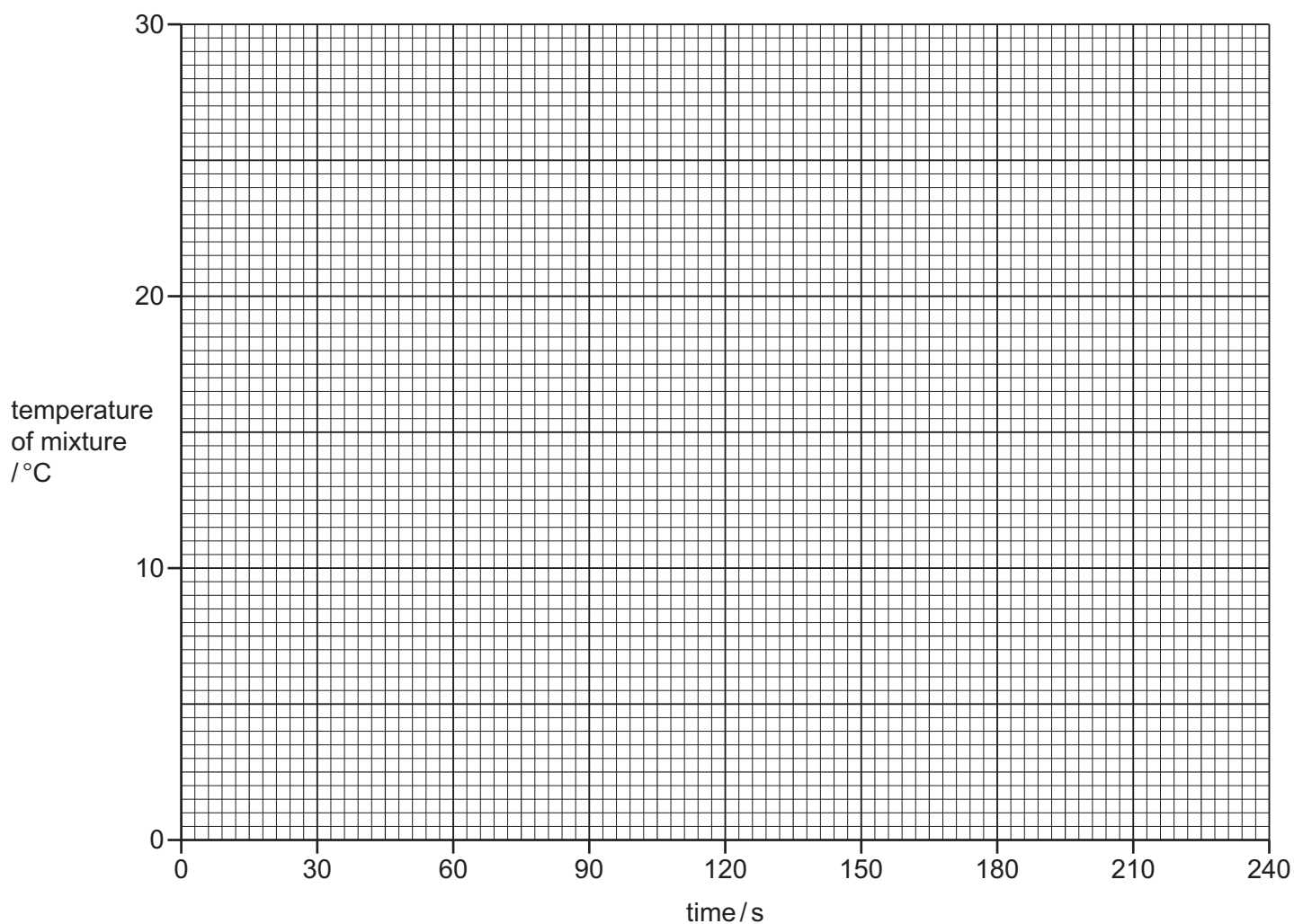
Experiment 1 was repeated using a new polystyrene cup and solid **O** instead of solid **N**.

(b) Use the thermometer diagrams to record the temperatures in the table.

time / s	0	30	60	90	120	150	180
thermometer diagram							
temperature of mixture / °C							

[2]

- (c) Plot the results for Experiments 1 and 2 on the grid. Draw **two** smooth line graphs. Clearly label your graphs.



[3]

- (d) (i) **From your graph**, deduce the time taken for the initial temperature of the solution in Experiment 2 to change by 3°C .

Show clearly **on the grid** how you worked out your answer.

..... s [3]

- (ii) Extend your graph for Experiment 1 to give the expected temperature of the mixture after 240 seconds.

..... $^{\circ}\text{C}$ [2]

- (e) Is the energy change in Experiment 2 exothermic or endothermic? Explain your answer.

.....
 [1]

- (f) State **two** possible sources of error in these experiments. Suggest **two** improvements to reduce each of these sources of error.

source of error 1

improvement 1

source of error 2

improvement 2

[4]

[Total: 17]

- 4 Potassium nitrate and ammonium chloride are two salts. The energy change when they each dissolve in water is endothermic.

Plan an experiment to show which of these two salts produces the larger endothermic energy change per gram.

Your answer should include:

- any measurements you would take and record
- how the results could be used to draw a conclusion.

You are provided with potassium nitrate and ammonium chloride, distilled water and common laboratory apparatus.

[6]

[6]

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- 2 A student investigated the temperature change when magnesium ribbon reacts with dilute sulfuric acid.

Five experiments were done.

Experiment 1

- Using a measuring cylinder, 20 cm³ of dilute sulfuric acid were poured into a boiling tube.
- A thermometer was used to measure the initial temperature of the acid.
- A 1 cm length of magnesium ribbon was added to the acid in the boiling tube.
- The acid and magnesium ribbon in the boiling tube were stirred continuously using a thermometer.
- The highest temperature reached by the mixture was measured.
- The boiling tube was rinsed out with distilled water.

Experiment 2

- Experiment 1 was repeated using a 2 cm length of magnesium ribbon instead of the 1 cm length.

Experiment 3

- Experiment 1 was repeated using a 3 cm length of magnesium ribbon instead of the 1 cm length.

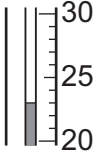
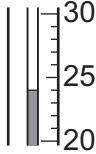
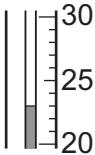
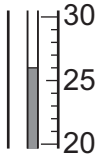
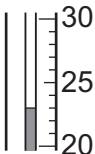
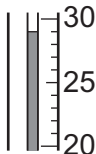
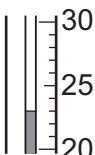
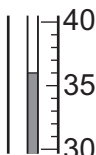
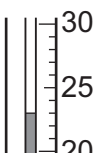
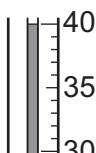
Experiment 4

- Experiment 1 was repeated using a 5 cm length of magnesium ribbon instead of the 1 cm length.

Experiment 5

- Experiment 1 was repeated using a 6 cm length of magnesium ribbon instead of the 1 cm length.

- (a) Use the information in the description of the experiments and the thermometer diagrams to complete the table.

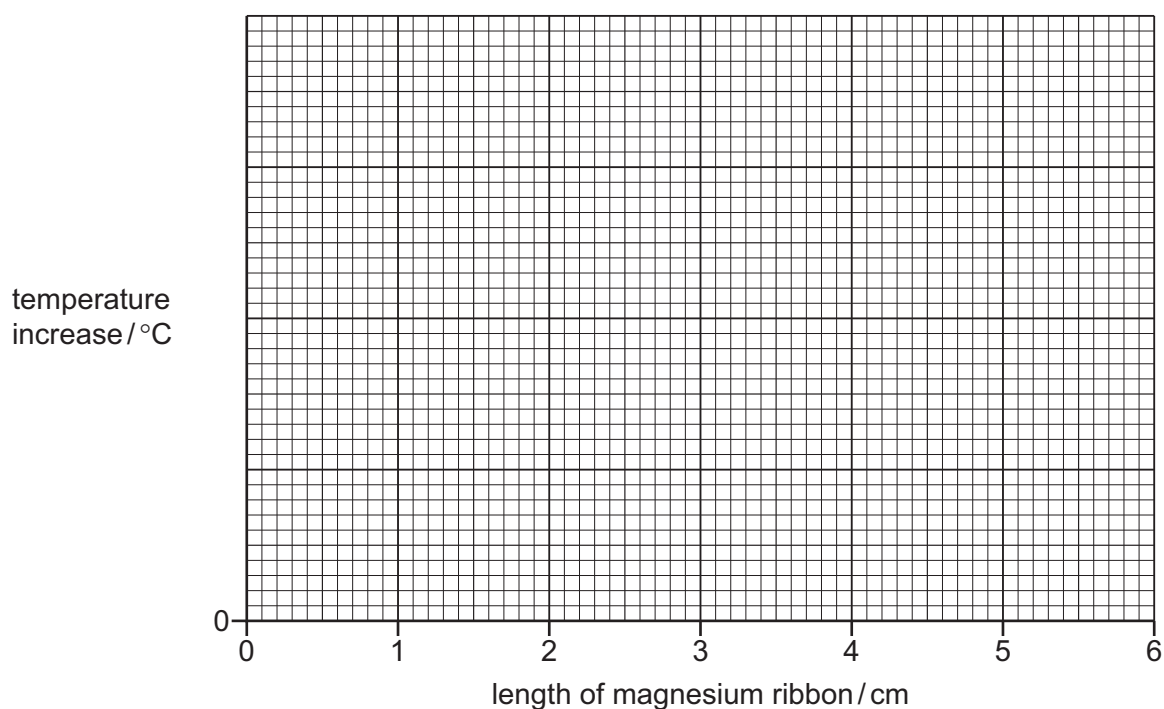
experiment	length of magnesium ribbon / cm	initial temperature		highest temperature		temperature increase / °C
		thermometer diagram	initial temperature of acid / °C	thermometer diagram	highest temperature of acid / °C	
1						
2						
3						
4						
5						

[4]

- (b) In which experiment, 1, 2, 3, 4 or 5, was the temperature increase the largest?

..... [1]

- (c) Add a suitable scale to the y-axis and plot the results from Experiments 1 to 5 on the grid. Draw a smooth line graph, making sure that your line passes through (0,0).



[5]

- (d) Explain why the graph line must pass through (0,0).

.....
 [1]

- (e) **From your graph**, deduce the temperature increase if Experiment 1 is repeated using a 4 cm length of magnesium ribbon.

Show clearly **on the grid** how you worked out your answer.

.....
 [3]

- (f) (i) Why would carrying out the experiment in a polystyrene cup rather than a boiling tube improve the accuracy of the results?

.....
 [1]

- (ii) Sketch **on the grid** the graph you would expect if the experiment was repeated using a polystyrene cup instead of a boiling tube. [1]

(g) The volume of dilute sulfuric acid could be measured with a 20 cm³ pipette.

(i) State **one** advantage of using a pipette rather than a measuring cylinder.

.....

..... [1]

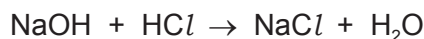
(ii) State **one** disadvantage of using a pipette rather than a measuring cylinder.

.....

..... [1]

[Total: 18]

- 2 A student investigated the temperature change when aqueous sodium hydroxide neutralises dilute hydrochloric acid. The equation for the reaction is shown.



Eight experiments were done.

Experiment 1

- A polystyrene cup was placed into a 250 cm³ beaker for support.
- Using a measuring cylinder, 5 cm³ of aqueous sodium hydroxide was poured into the polystyrene cup.
- Using a measuring cylinder, 45 cm³ of dilute hydrochloric acid was poured into the polystyrene cup.
- The mixture was stirred and the maximum temperature reached was measured using a thermometer.
- The polystyrene cup was rinsed with distilled water.

Experiment 2

- Experiment 1 was repeated using 10 cm³ of aqueous sodium hydroxide and 40 cm³ of dilute hydrochloric acid.

Experiment 3

- Experiment 1 was repeated using 15 cm³ of aqueous sodium hydroxide and 35 cm³ of dilute hydrochloric acid.

Experiment 4

- Experiment 1 was repeated using 20 cm³ of aqueous sodium hydroxide and 30 cm³ of dilute hydrochloric acid.

Experiment 5

- Experiment 1 was repeated using 30 cm³ of aqueous sodium hydroxide and 20 cm³ of dilute hydrochloric acid.

Experiment 6

- Experiment 1 was repeated using 35 cm³ of aqueous sodium hydroxide and 15 cm³ of dilute hydrochloric acid.

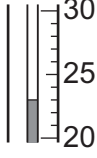
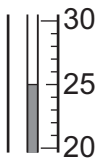
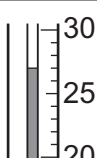
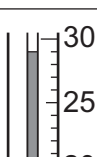
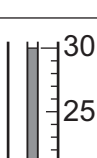
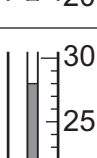
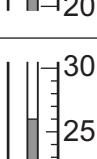
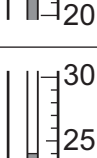
Experiment 7

- Experiment 1 was repeated using 40 cm³ of aqueous sodium hydroxide and 10 cm³ of dilute hydrochloric acid.

Experiment 8

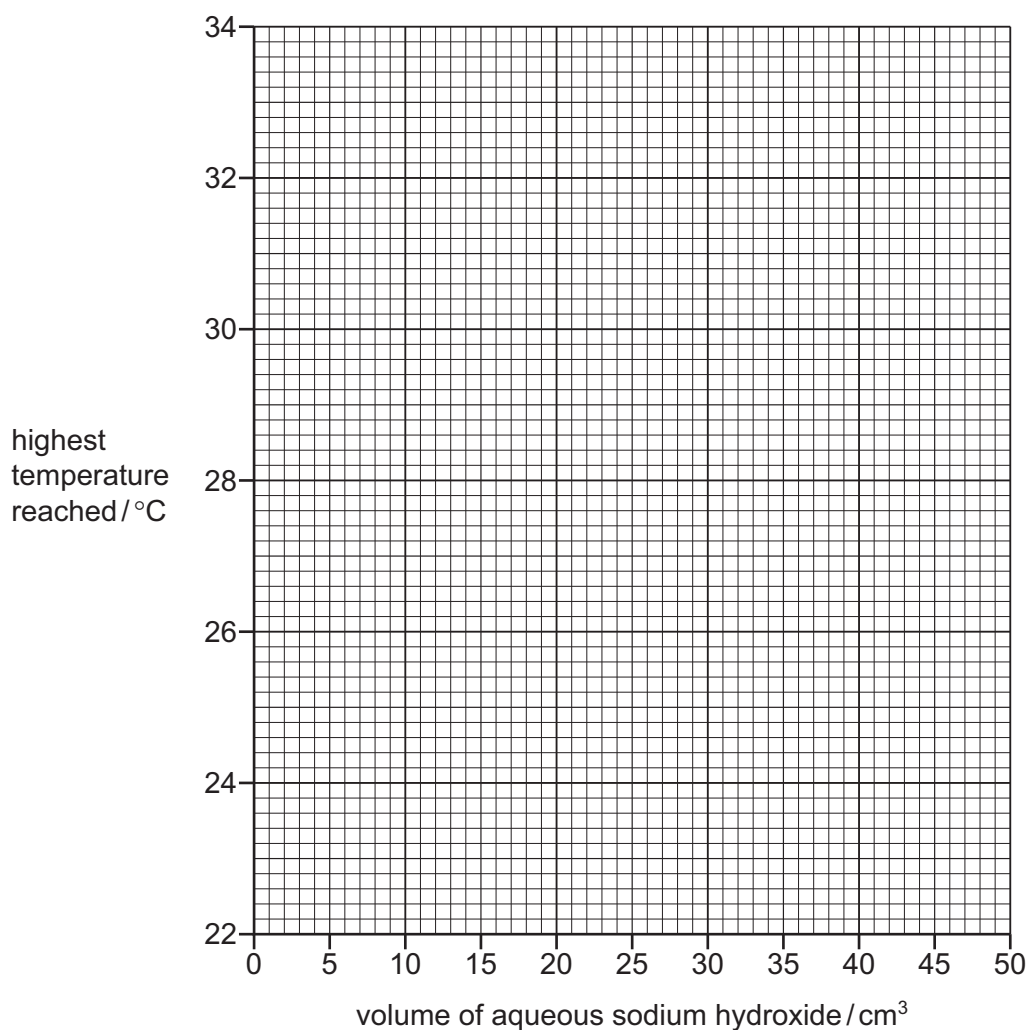
- Experiment 1 was repeated using 45 cm³ of aqueous sodium hydroxide and 5 cm³ of dilute hydrochloric acid.

- (a) Use the information in the description of the experiments and the thermometer diagrams to complete the table.

experiment	volume of aqueous sodium hydroxide / cm ³	volume of dilute hydrochloric acid / cm ³	thermometer diagram	highest temperature reached / °C
1	5			
2	10			
3	15			
4	20			
5	30			
6	35			
7	40			
8	45			

[4]

- (b) Plot the results from Experiments 1 to 8 on the grid. Draw **two** straight lines through the points. Extend your straight lines so that they cross.



[4]

- (c) The point on the graph where the two straight lines cross is where all of the aqueous sodium hydroxide reacts with all of the dilute hydrochloric acid to form a neutral solution.

- (i) **Use your graph** to deduce the volume of aqueous sodium hydroxide and the volume of dilute hydrochloric acid that react together to produce a neutral solution. Show your working **on the grid**.

volume of aqueous sodium hydroxide = cm³

volume of dilute hydrochloric acid = cm³

[3]

- (ii) **Use your graph** to determine the highest temperature reached if the volumes in (c)(i) were mixed together.

highest temperature reached = [2]

- (iii) Which solution, aqueous sodium hydroxide or dilute hydrochloric acid, was the most concentrated?

Use your answer to (c)(i) to explain why.

most concentrated solution

explanation

..... [1]

- (d) **On the graph**, sketch the lines you would expect to obtain if a copper can was used instead of a polystyrene cup. [2]

- (e) Give **one** advantage and **one** disadvantage of using a burette, instead of a measuring cylinder, to add the dilute hydrochloric acid directly into the polystyrene cup.

advantage

.....

disadvantage

.....

[2]

- (f) How could the reliability of the results of this investigation be checked?

.....

..... [1]

[Total: 19]

- 2 A student investigated the temperature decrease when sodium hydrogencarbonate reacts with dilute hydrochloric acid.

The student did six experiments.

Experiment 1

- Using a measuring cylinder, 25 cm³ of dilute hydrochloric acid was poured into a conical flask.
- The initial temperature of the acid was measured using a thermometer.
- 1 g of sodium hydrogencarbonate was added to the conical flask. At the same time a stop-clock was started.
- The acid and sodium hydrogencarbonate mixture in the conical flask was stirred continuously using the thermometer.
- The temperature of the mixture after 1 minute was measured.
- The conical flask was rinsed with distilled water.

Experiment 2

- Experiment 1 was repeated using 2 g of sodium hydrogencarbonate instead of 1 g.

Experiment 3

- Experiment 1 was repeated using 3 g of sodium hydrogencarbonate instead of 1 g.

Experiment 4

- Experiment 1 was repeated using 5 g of sodium hydrogencarbonate instead of 1 g.

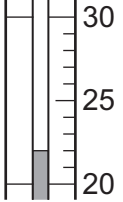
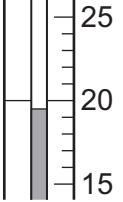
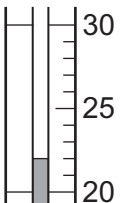
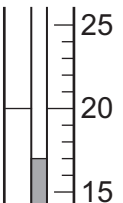
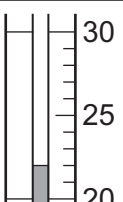
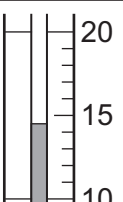
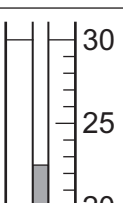
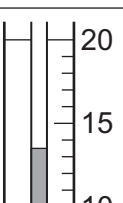
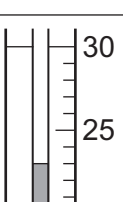
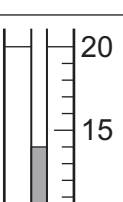
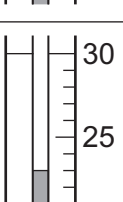
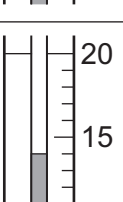
Experiment 5

- Experiment 1 was repeated using 6 g of sodium hydrogencarbonate instead of 1 g.

Experiment 6

- Experiment 1 was repeated using 7 g of sodium hydrogencarbonate instead of 1 g.

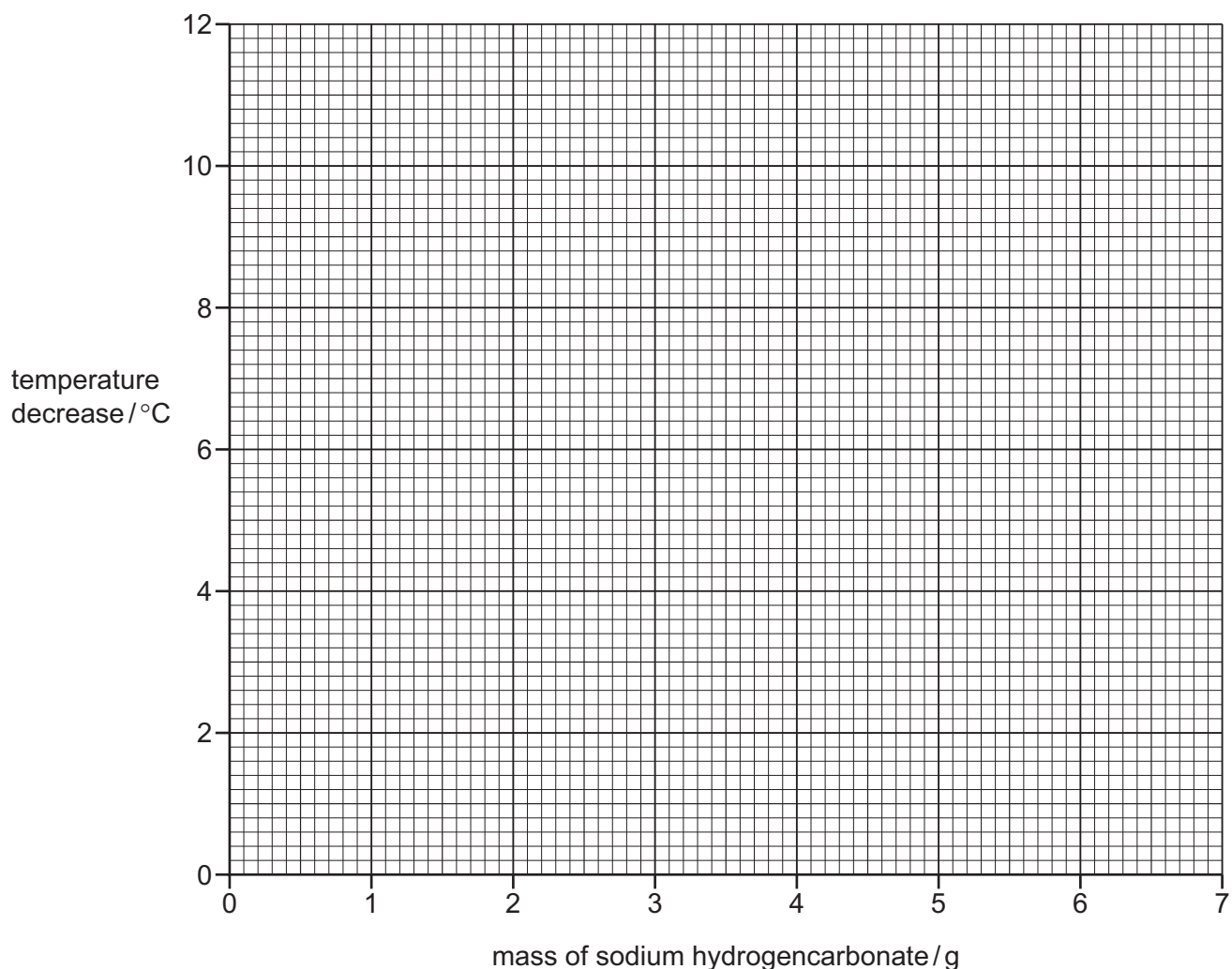
(a) Use the thermometer diagrams to complete the table and calculate the temperature decreases.

experiment	mass of sodium hydrogencarbonate /g	thermometer diagram	initial temperature of acid /°C	thermometer diagram	temperature after 1 minute /°C	temperature decrease /°C
1	1					
2	2					
3	3					
4	5					
5	6					
6	7					

[4]

- (b) Plot the results from Experiments 1 to 6 on the grid.

Draw **two** best-fit straight lines through your points. The first straight line should be for the first three points and must pass through (0,0). The second straight line should be for the last three points and must be horizontal. Extend your straight lines so that they meet each other.



[4]

- (c) (i) **From your graph**, determine the temperature decrease and mass of sodium hydrogencarbonate where your two straight lines meet. Include appropriate units in your answer.

Show clearly **on the grid** how you worked out your answer.

temperature decrease =

mass of sodium hydrogencarbonate =

[3]

- (ii) Explain why the temperature decrease becomes constant for high masses of sodium hydrogencarbonate.

.....
 [1]

- (d) The investigation was repeated with dilute hydrochloric acid of half the concentration, but the same volume.

Sketch **on the grid** the graph you would expect to obtain.

Label your line **D**.

[2]

- (e) Suggest **two** changes that could be made to the apparatus that would improve the accuracy of the results. For each change explain why it would improve the accuracy of the results.

change 1

explanation 1

.....

change 2

explanation 2

.....

[4]

[Total: 18]

- 2 A student investigated the temperature change when zinc reacted with two different aqueous solutions of copper(II) sulfate, solution **Q** and solution **R**.

Two experiments were done.

(a) *Experiment 1*

- A polystyrene cup was placed in a 250 cm³ beaker for support.
- Using a measuring cylinder, 25 cm³ of solution **Q** was poured into the polystyrene cup.
- Using a thermometer, the initial temperature of solution **Q** was measured.
- 3 g of zinc powder was added to the polystyrene cup. At the same time a stop-clock was started.
- Using the thermometer, the mixture in the polystyrene cup was continually stirred and the temperature measured every 30 seconds.

initial temperature in Experiment 1	23 °C
-------------------------------------	-------

Use the thermometer diagrams and the initial temperature to complete the table.
Calculate the temperature changes using the equation:

$$\text{temperature change} = \text{temperature} - \text{initial temperature}$$

time/s	30	60	90	120	150	180	210	240
thermometer diagram								
temperature/°C								
temperature change/°C								

[3]

(b) Experiment 2

- The polystyrene cup was washed out with distilled water.
- Experiment 1 was repeated using solution **R** instead of solution **Q**.

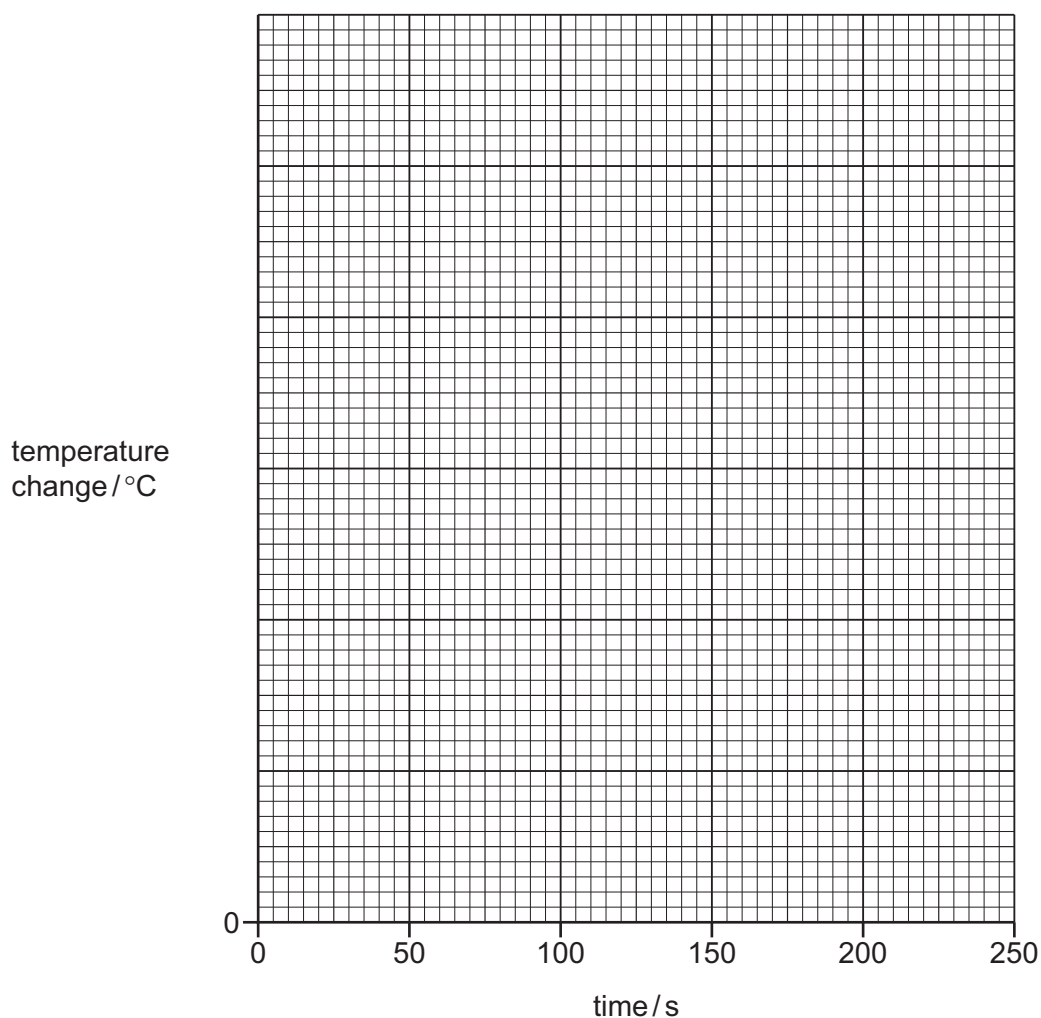
initial temperature in Experiment 2	24 °C
-------------------------------------	-------

Use the thermometer diagrams and the initial temperature to complete the table.

time / s	30	60	90	120	150	180	210	240
thermometer diagram								
temperature / °C								
temperature change / °C								

[3]

- (c)** Complete a suitable scale on the y-axis and plot the results from Experiment 1 and Experiment 2 on the grid. Draw two curves of best fit. Both curves must start at (0,0). Label your curves.



[5]

- (d) From your graph, deduce the temperature change at 110 seconds in Experiment 1.

Show clearly **on the grid** how you worked out your answer.

..... °C
[2]

- (e) Predict the temperature of the solution in Experiment 2 after 5 hours. Explain your answer.

.....
..... [2]

- (f) (i) Suggest why the experiments were done in a polystyrene cup rather than a glass beaker.

..... [1]

- (ii) Describe how the results would be different if a glass beaker is used in place of the polystyrene cup.

..... [1]

- (g) Suggest **one** change that could be made to the apparatus that would improve the accuracy of the results. Explain why this change would improve the accuracy of the results.

change

explanation

.....
[2]

[Total: 19]

- 2 A student investigated the temperature change when anhydrous lithium chloride dissolves in water.

The student did six experiments.

(a) *Experiment 1*

- Using a measuring cylinder, 30 cm³ of distilled water was poured into a 100 cm³ beaker.
- The initial temperature of the water was measured using a thermometer.
- 1.0 g of anhydrous lithium chloride was added to the water in the beaker. At the same time a timer was started.
- The water and lithium chloride mixture was continually stirred using a thermometer.
- The temperature of the mixture was measured after 30 seconds.
- The beaker was rinsed with distilled water.

Experiment 2

- Experiment 1 was repeated using 1.5 g of anhydrous lithium chloride instead of the 1.0 g of anhydrous lithium chloride.

Experiment 3

- Experiment 1 was repeated using 2.0 g of anhydrous lithium chloride instead of the 1.0 g of anhydrous lithium chloride.

Experiment 4

- Experiment 1 was repeated using 2.5 g of anhydrous lithium chloride instead of the 1.0 g of anhydrous lithium chloride.

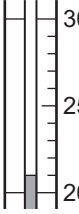
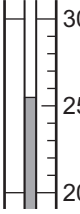
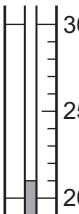
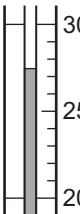
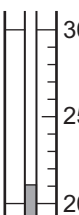
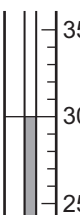
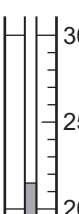
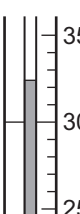
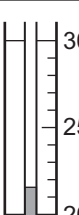
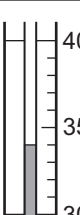
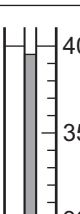
Experiment 5

- Experiment 1 was repeated using 3.0 g of anhydrous lithium chloride instead of the 1.0 g of anhydrous lithium chloride.

Experiment 6

- Experiment 1 was repeated using 4.0 g of anhydrous lithium chloride instead of the 1.0 g of anhydrous lithium chloride.

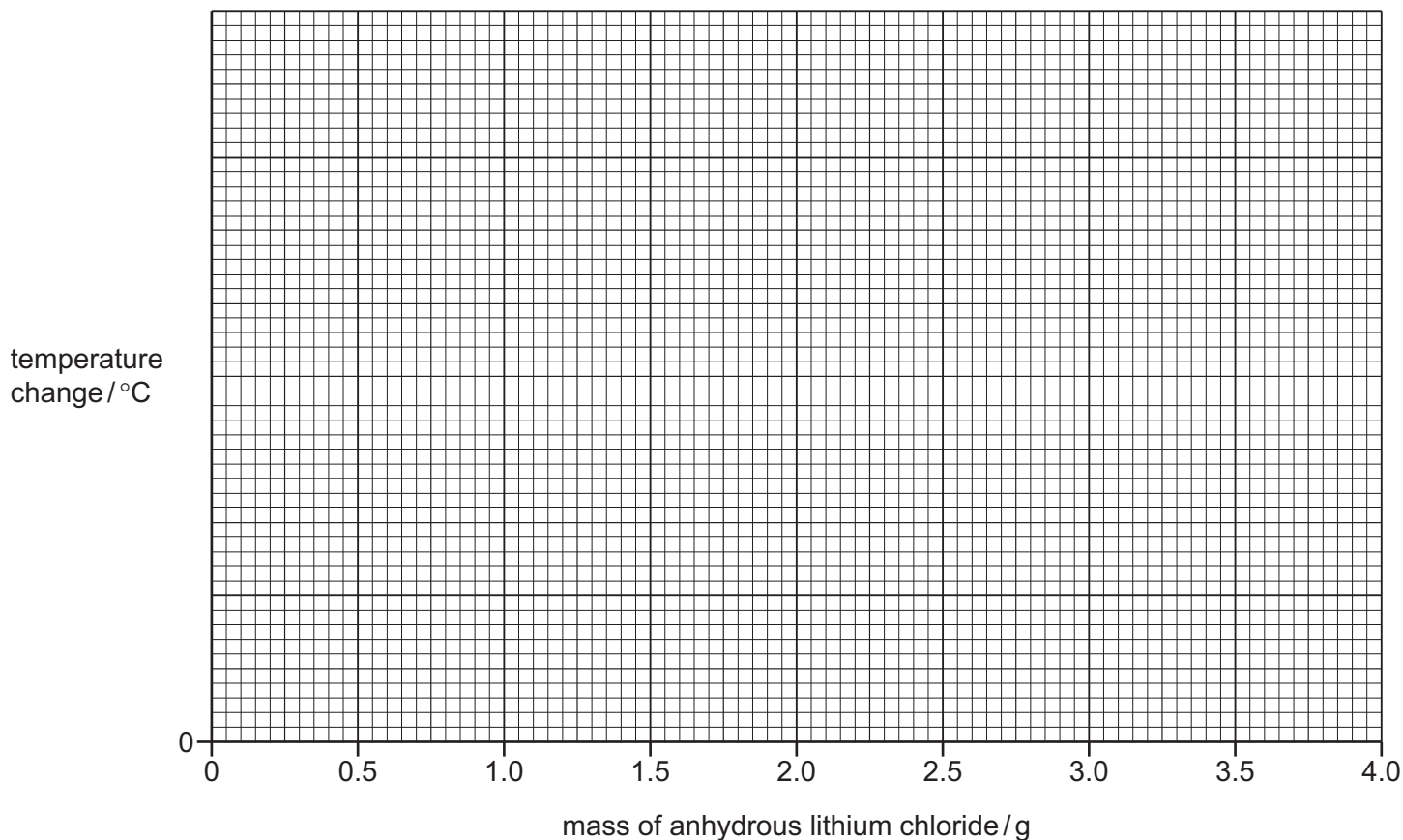
Use the thermometer diagrams to complete the table and calculate the temperature changes.

experiment	mass of anhydrous lithium chloride /g	initial		after 30 seconds		temperature change /°C
		thermometer diagram	temperature /°C	thermometer diagram	temperature /°C	
1	1.0					
2	1.5					
3	2.0					
4	2.5					
5	3.0					
6	4.0					

[5]

- (b) Complete a suitable scale on the y-axis and plot the results from Experiments 1 to 6 on the grid.

Draw a straight line of best fit through your points. The straight line must pass through (0,0).



[5]

- (c) **From your graph**, deduce the temperature change when 3.2 g of anhydrous lithium chloride is dissolved in 30 cm³ of distilled water.

Show clearly **on the grid** how you worked out your answer.

temperature change = °C [2]

- (d) Estimate the temperature change if Experiment 6 is repeated using 60 cm³ of water instead of 30 cm³ of water. Give a reason for your answer.

.....

 [2]

- (e) Suggest **two** changes that could be made to the apparatus to improve the accuracy of the results. For each change explain why it improves the accuracy of the results.

change 1

explanation 1

.....

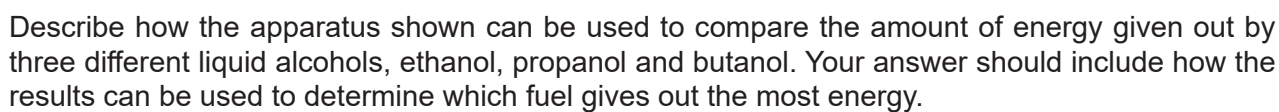
change 2

explanation 2

.....

[4]

[Total: 18]



..... [6]

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